THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a trapezium with parallel sides 8 cm and 6 cm and height 5 cm is: 1

(a) 35 cm² **(b)** 120 cm² **(c)** 40 cm² **(d)** 70 cm²

2. A polyhedron has 6 faces and 8 vertices. By Euler's formula, its number of edges is: 1

(a) 24 **(b)** 14 **(c)** 12 **(d)** 13

3. The mean of 6 numbers is 15. Their sum is: 1

(a) 96 **(b)** 90 **(c)** 16 **(d)** 21

4. In a calendar, the four dates 3, 4, 10 and 11 form a 2×2 block. Their sum is: 1

(a) 28 **(b)** 26 **(c)** 30 **(d)** 20

5. The area of a rhombus with diagonals 10 cm and 8 cm is: 1

(a) 80 cm² **(b)** 18 cm² **(c)** 40 cm² **(d)** 36 cm²

6. The sum of the lengths of all the edges of a cube of side 3 cm is (in cm): 1

(a) 24 **(b)** 36 **(c)** 18 **(d)** 12

7. In a dot plot each dot represents: 1

(a) the mean **(b)** the median **(c)** ten observations **(d)** one observation

8. In the expression $5x + 7$, the coefficient of x is: 1

(a) x **(b)** 12 **(c)** 7 **(d)** 5

9. A square has a perimeter of 20 cm. Its area is: 1

(a) 20 cm² **(b)** 25 cm² **(c)** 50 cm² **(d)** 30 cm²

10. Euler's formula for a polyhedron with F faces, V vertices and E edges is: 1

(a) $F - V + E = 2$ **(b)** $F + V - E = 2$ **(c)** $F + V + E = 2$ **(d)** $F + V = E$

11. The mean of 11, 15, 19, 7, 13 is: 1

(a) 14 **(b)** 13 **(c)** 26 **(d)** 12

12. The expression for '4 more than 3 times a number x' is: 1

(a) $3x + 4$ **(b)** $3(x + 4)$ **(c)** $4x + 3$ **(d)** $3 + 7x$

13. The area of a rectangle 15 cm long and 8 cm broad is: 1

(a) 23 cm² **(b)** 120 cm² **(c)** 46 cm² **(d)** 135 cm²

14. In the Sierpinski triangle construction, the number of small shaded triangles at stage 2 (stage 0 has one) is: 1

(a) 9 **(b)** 8 **(c)** 12 **(d)** 6

15. The median of 17, 24, 35, 16, 29, 30, 14 is: 1

(a) 16 **(b)** 14 **(c)** 24 **(d)** 23.57

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the median of 60, 53, 56, 32, 38, 5, 33. 2

17. Find the value of $3x + 2y$ when $x = 4$ and $y = 5$. 2

18. Find the area of a triangle with base 18 cm and height 10 cm. 2

19. Find the total length of all edges of a cube of side 4 cm. 2

20. Find the range and the mode of 12, 10, 9, 13, 10, 10, 15. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the area of a trapezium whose parallel sides are 13 cm and 7 cm and the distance between them is 9 cm. 3

22. A cuboid measures 5 cm by 3 cm by 2 cm. Find the total length of its edges and state its number of faces and vertices. 3

23. The temperatures (°C) at 6 am, 9 am, 12 noon and 3 pm were 14, 18, 25 and 28. Between which two consecutive times was the rise greatest, and what was the rise? 3

24. Find the sum of 23 and its reverse 32. Show that it is a multiple of 11. 3

OR

A rectangular floor measures 8 m by 5 m. Find its area and the cost of tiling it at ₹ 40 per m². 3

25. State the number of faces, vertices and edges of a square pyramid and verify Euler's formula. 3

OR

The marks of 6 students are 24, 81, 82, 26, 78, 27. Find the mean mark and the number of students who scored above the mean. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. In the Sierpinski triangle, stage 0 has 1 shaded triangle. Fill in the number of shaded triangles at stages 1, 2 and 3. By what factor does the count grow at each stage? 5

OR

The mean of 5 numbers is 20. Four of them are 12, 18, 25, 30. Find the fifth number. 5

27. In a number pyramid each block is the sum of the two blocks below it. The bottom row is 2, 5, 3. Find the middle row and the top block. 5

OR

The diagonals of a rhombus are 16 cm and 12 cm. Find its area, and the area of each of the four triangles into which the diagonals divide it. 5

END OF QUESTION PAPER — Total: 50 Marks